Seunghyun Yoo

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Research Interests

Mechanistic interpretability, technical AI safety, and multi-agent systems

Education

Purdue University Aug. 2024 – Expected Dec. 2027
B.S. in Artificial Intelligence; Minor in Mathematics *West Lafayette, IN*

Publications

- **Seunghyun Yoo**. *White Box Evidence Packages for Policy Audit Reports*. Second Workshop on Technical AI Governance Research (TAIGR), ICML 2026. Sole author. [Paper] [Poster]
- Meliksah Yorulmazlar, Justice Sefas, **Seunghyun Yoo**, and Lulu Shao. *Feature Evolution in Diffusion Language Models*. Non-Autoregressive Language Models Workshop, COLM 2026; poster. [Paper] [Poster] [Live demo]

Research Experience

Research Fellow Sep. 2026 – Present
Supervised Program for Alignment Research; Mentors: Yuxiao Li and Di Wu *Remote*

- Studying how latent influence propagates across interacting agents, using activation steering and representation-level analysis to measure behavioral and internal-state shifts over extended conversations.

Undergraduate Researcher (Team Lead) Jun. 2025 – Present
Governance and Responsible AI Lab (GRAIL) / Purdue AI Forge *Purdue University*

- Designed and implemented a policy audit evaluation spanning 60 AGORA cases, 600 reports, and ten evidence conditions. Demonstrated that audit reports can cite irrelevant internal model evidence despite appearing plausible; authored the resulting TAIGR paper.
- Built the policy representation analysis from corpus preparation through SAE extraction, statistical testing, and causal interventions in Gemma 2 2B. Analyzed 2,438 records and identified 35 layer 24 features stable across five permutation seeds.
- Developed AGORA-RAGTruth, a policy hallucination benchmark with manual span annotations and governance tags, and designed mechanistic analyses separating retrieved context use from parametric knowledge.
- Annotated and validated AGORA policy documents and redesigned the Explore-AGORA graph schema to connect policy passages with risks, strategies, and institutional metadata.

Research Fellow Feb. 2026 – Jun. 2026
Algoverse AI Safety Fellowship; Supervisor: Justice Sefas *Remote*

- Designed and implemented analyses of SAE features in Dream-v0-Base-7B across six layers and 64 denoising steps, including activation statistics, feature interpretation, and token commitment visualizations.
- Identified recurring patterns of masked token tracking, activation near token commitment, and selective responses to unmasked tokens. Found limited evidence for clean early or late feature stages in the studied setup.
- Analyzed token unmasking across mathematical, natural language, and code reasoning prompts. The resulting paper, *Feature Evolution in Diffusion Language Models*, was accepted to the Non-Autoregressive Language Models Workshop at COLM 2026.

Hackathon Projects

Ethogram [Live demo] [Code] Oct. 2026
AI Village × Grove Research AI Swarm Dynamics Hackathon

- Built a pipeline that turns multi-agent logs into a plain-language behavior dictionary, a storyline of phases and turning points, and an interactive behavior atlas, with every claim linked to the events behind it.
- Validated behaviors with two independent LLM judges and blind detection by a different model; two-judge agreement reached $\kappa = 0.89$ and 0.86 on two real records of 20,932 and 67,969 events.

MARGIN Arena: Agents Under Resource Limits [Paper] [Code] Sep. 2026
AI Incident Response Hackathon; Apart Research *Remote*

- Designed and published a Python environment for studying agent behavior under five independently configurable limits on tokens, time, verification calls, tool calls, and sessions.
- Integrated Inspect and ControlArena tasks with original scoring, preserved execution records, and optional activation capture for interpretability. Validated resource accounting, stopping behavior, and analysis interfaces with 50 software tests.

Denoising Trajectories as Early-Warning Signals for Diffusion Language Model Control [Paper] [Code] Mar. 2026
AI Control Hackathon; Apart Research / Redwood Research *Remote*

- Built a monitoring and steering system for LLaDA-8B-Instruct using pretrained SAEs, with executable shell benchmarks and tests under prompt and payload variation.
- Detected attacks at mean denoising step 9.60 versus 63.00 for output checks on the main benchmark; early detection persisted under variation (10.53 versus 63.00).

Teaching

AI Study Leader Sep. 2025 – Present
LikeLion US / OZ Lab *West Lafayette, IN*

- Teach weekly sessions on LLM From Scratch, MiniArena, and Interpretability for Purdue's Korean student community.

Selected Posters and Presentations

- *Co-CoT: A Prompt-Based Framework for Collaborative Chain-of-Thought Reasoning*. RISE AI Conference, Oct. 2025. Poster / presentation; first author.
- *Factually Grounded and Human-Aligned RAG Systems for AI Governance and Policy*. Purdue URC, 2025. Poster.
- Internal GRAIL presentations: *Explore-AGORA 2* (Feb. 2026) and *DPO + Explore-AGORA* (Nov. 2025).

Honors and Recognition

Purdue PLaCE Student Spotlight, 2024. Travel Grant Recipient, RISE AI Conference, University of Notre Dame.

Technical Skills

Programming and ML Python, PyTorch, Hugging Face Transformers, Git.
Language models RAG, diffusion models, model evaluation.
Interpretability Sparse autoencoders, activation patching, probing, TransformerLens, NNsight, NDIF.

Activities

AI Safety Purdue and ML@Purdue, general member; Purdue eSports, Overwatch Junior Varsity.